$$Y\frac{\partial u}{\partial x}+X\frac{\partial u}{\partial y}=0$$

$$\frac{dx}{Y}=\frac{dy}{X}$$

$$X\,dx=Y\,dy$$

$$\frac{X^2}{2}=\frac{Y^2}{2}+C_1$$

$$2C_1=C_2=X^2-Y^2$$

$$u=\varphi(X^2-Y^2)$$

$$Y\frac{\partial u}{\partial x}+X\frac{\partial u}{\partial y}=Xu$$

$$\frac{dx}{Y}=\frac{dy}{X}=\frac{du}{X}$$

$$C_1=X^2-Y^2$$

$$\frac{dy}{X}=\frac{du}{X}$$

$$y=u+C_2$$

$$C_2=y-u$$

$$u=Y-C_2=Y-\varphi(X^2-Y^2)$$

$y=0$ $\quad aX^2=-\varphi(X^2)$

$u=aX^2$

$$\varphi(X^2)=-aX^2$$

$$\varphi(X^2-Y^2)=-a(X^2+Y^2)$$

$$u=Y+a(X^2+Y^2)$$

$$\alpha^2 \frac{\partial^2 T}{\partial x^2} = \frac{\partial T}{\partial t}$$

$$\alpha^2 \frac{\partial^2 T}{\partial x^2} = \frac{\partial T}{\partial t}$$

$$0 < x < L$$

נפריד משתנים

$$T = X(x)\,T(t)$$

$$\alpha^2 \frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha^2}$$

$$T_{(t)} \frac{d^2X}{dx^2} = \frac{1}{\alpha^2} X_{(x)} \frac{dT}{dt}$$

$$\frac{1}{X} \frac{d^2X}{dx^2} = \frac{1}{\alpha^2} \frac{1}{T} \frac{dT}{dt} = -\lambda$$

$$\frac{1}{T\alpha^2} \frac{dT}{dt} = -\lambda \Rightarrow \frac{dT}{dt} + \lambda\alpha^2 T = 0$$

נסמן $\lambda \equiv k^2$

$$T' + k^2\alpha^2 T = 0$$

$$\left(Te^{+\alpha^2 k^2 t}\right)' = 0$$

$$Te^{+\alpha^2 k^2 t} = C$$

$$T = e^{-\alpha^2 k^2 t}$$

$$\frac{d^2X}{dx^2} + k^2 X = 0 \qquad X_{(x)} = A_k \cos kx + B_k \sin kx$$

$$X_{(0)} = 0 = A_k$$

משיקולי גבולות

$$X_{(L)} = B_k \sin kL = 0 \Rightarrow kL = n\pi$$

$$T(x,t) = \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L} e^{-\alpha^2 k^2 t} + B_0 e^{-\alpha^2 k^2 t}$$

$$T(x,0) = f(x)$$

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L} + B_0$$

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx$$

$$B_0 = \frac{2}{L}\int_0^L f(x)\,dx$$

$$T(L,t) = T_2 \qquad T(0,t) = T_1$$

~~$Y_p(x,t) = C_1 + C_2 x + C_3(x^2 + \alpha^2 t)$~~

$$Y_p(x,t) = C_1 + C_2 x + C_3(x^2 + 2\alpha^2 t)$$

$$W_H(x,t) = \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L} e^{-\alpha^2 k^2 t} + B_0 e^{-\alpha^2 k^2 t}$$

$$U(x,t) = Y_p(x,t) + W_H(x,t)$$

$$U(0,t) = Y_p(0,t) + W_H(0,t) = T_1 = C_1 + 2C_3\alpha^2 t \qquad C_3 = 0$$

$$U(L,t) = Y_p(L,t) + W_H(L,t) = T_2 = C_1 + C_2 L + C_3(L^2 + 2\alpha^2 t)$$

$$T_1 = C_1$$

$$T_2 = T_1 + C_2 L \Rightarrow C_2 = \frac{T_2 - T_1}{L}$$

$$T(x,t) = T_1 + \frac{T_2 - T_1}{L} x + \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L} e^{-\alpha^2 k^2 t}$$

$$T(x,t=0) = T_1 + (T_2-T_1)\theta(x-\tfrac{L}{2}) = T_1 + \frac{T_2-T_1}{L}x + B_0 + \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L}$$

$$(T_2-T_1)\theta(x-\tfrac{L}{2}) = f(x) = \frac{T_2-T_1}{L}x + B_0$$

$$u(x,0) = B_0 + \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L} = T_1 + \frac{T_2-T_1}{L}x$$

$$T(x,t=0) = f(x) = T_1 + \frac{T_2-T_1}{L}x + \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L}$$

$$f(x) - T_1 - \frac{T_2-T_1}{L}x = \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L}$$

$$B_n = \frac{2}{L}\int_0^L \left(f(y) - T_1 - \frac{T_2-T_1}{L}y\right)\sin\frac{n\pi y}{L}\,dy =$$

$$= \frac{2}{L}\int_0^L \left(T_1 + (T_2-T_1)\theta(x-\tfrac{L}{2}) - T_1 - \frac{T_2-T_1}{L}y\right)\sin\frac{n\pi y}{L}\,dy =$$

$$= \frac{2}{L}\int_{\frac{L}{2}}^{L}(T_2-T_1)\sin\frac{n\pi y}{L}\,dy - \frac{2}{L}\int_0^L \frac{T_2-T_1}{L}y\sin\frac{n\pi y}{L}\,dy =$$

$$= \frac{2}{L}(T_2-T_1)\left(-\frac{L}{n\pi}\cos\frac{n\pi y}{L}\Big|_{\frac{L}{2}}^{L}\right) - \frac{2(T_2-T_1)}{L^2}\left(-\frac{L}{n\pi}y\cos\frac{n\pi y}{L}\Big|_0^L + \frac{L}{n\pi}\int_0^L \cos\frac{n\pi y}{L}\,dy\right)$$

$$= -\frac{2(T_2-T_1)}{n\pi}\left(\cos n\pi - \cos\frac{n\pi}{2}\right) + \frac{2(T_2-T_1)}{L}\left(\frac{L}{n\pi}\cos n\pi\right)$$

$$\frac{2}{n\pi}(T_2-T_1)\left(\cos\frac{n\pi}{2} - \cos n\pi\right) + \frac{2(T_2-T_1)}{n\pi}\cos n\pi =$$

$$= -\frac{2(T_2-T_1)}{n\pi}(-1)^n + \frac{2(T_2-T_1)}{n\pi}(-1)^n = \frac{2(T_2-T_1)}{n\pi}\left((-1)^n - (-1)^n\right)$$

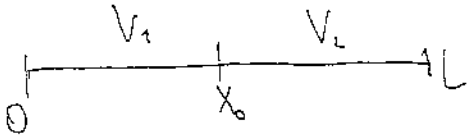
$0<x<L$

$k>0$

$Lu = u'' - k^2 u$

$$L_x G(x,x_0) = -\delta(x-x_0)$$

$$G(x_0,0) = G(x_0,L) = 0$$



$$0<x<x_0: \quad \frac{d^2 V_1}{dx^2} - k^2 V_1 = 0$$

$$x_0<x<L: \quad \frac{d^2 V_2}{dx^2} - k^2 V_2 = 0$$

$$V_i(x) = c_i e^{-kx} + d_i e^{kx}$$

~~$V_1(0) = 0 = c_1 + c_2 \Rightarrow c_1 = -c_2$~~

~~$V_1(0) = 0 = c_1 + d_1 \Rightarrow c_1 = -d_1$~~

~~$V_1(x) = c_1 e^{-kx} + c_1 e^{kx}$~~

~~$V_2(L) = 0 = c_2 e^{-kL} + d_2 e^{kL}$~~

מכיוון ש $\sinh$ ו $\cosh$ בנויות מהאקספוננטים

$$V_i(x) = a_i \sinh kx + b_i \cosh kx$$

$$V_1(0) = a_1 \underbrace{\sinh 0}_{0} + b_1 \underbrace{\cosh 0}_{1} = 0 \qquad x<x_0$$

$$\Rightarrow b_1 = 0$$

$$V_1(x) = a_1 \sinh kx$$

$a_2 = \cosh kL \qquad b_2 = -\sinh kL$

$V_1(x) = a_1 \sinh kx$

$V_2(x) = \cosh kL \sinh kx - \sinh kL \cosh kx = \sinh k(x-L)$



$$G = -\frac{1}{W} \begin{cases} V_1(x) V_2(x_0) & x < x_0 \\ V_1(x_0) V_2(x) & x > x_0 \end{cases}$$

$$W = \begin{vmatrix} \sinh kx & \sinh k(x-L) \\ k\cosh kx & k\cosh k(x-L) \end{vmatrix} = k\sinh kx \cosh k(x-L) - k\sinh k(x-L)\cosh kx$$

$$= k\sinh k(x - (x-L)) = k\sinh kL$$

$$G = -\frac{1}{k\sinh kL} \begin{cases} \sinh kx \, \sinh k(x_0 - L) \\ \sinh kx_0 \, \sinh k(x-L) \end{cases}$$

$$Lu + \lambda r u = 0 \qquad u(1) = 1 \qquad u(0) = 0$$

$$Lu = -\lambda r u$$

$$u(x) = \frac{\beta(x)}{\alpha(x)} + \lambda \int G(x,y) u(y) \, dy$$

$$\int (u \underbrace{LG}_{-\delta(x-y)} - G \underbrace{Lu}_{-\frac{1}{p}f(y)}) \, dy = u(x) + \int f(y) G(x,y) \, dy$$

$$\int [u(\tfrac{d^2}{dy^2}G - k^2 G) - G(\tfrac{d^2u}{dy^2} - k^2 u)] \, dy =$$

$$= u\frac{dG}{dy} - \int \frac{du}{dy}\frac{dG}{dy}\,dy - k^2\int uG\,dy - \left(G\frac{du}{dy} - \int \frac{dG}{dy}\frac{du}{dy}\,dy\right) +$$

$$+ k^2 \int GU \, dy = u\frac{dG}{dy} - G\frac{du}{dy}$$

$u(0)=0$, $u(2)=1$

$$u(x) = u(2)\frac{dG}{dy}\Big|_{y=2} - u(0)\frac{dG}{dy}\Big|_{y=0} - G|_{y=2}\frac{du}{dy}\Big|_{y=2} + G|_{y=0}\frac{du}{dy}\Big|_{y=0} + p\int_0^{2} G(x,y)\,dy$$

$$u(x) = \frac{dG}{dy}\Big|_{y=L} + p\int_0^{L} G(x,y)\,dy$$

$$G|_{y=L} = \frac{\sinh kx \sinh k(y-L)}{k\sinh kL}\Big|_{y=L}$$

$$\frac{dG}{dy}\Big|_{y=L} = \frac{k\sinh kx \cosh k(y-L)}{k\sinh kL} = \frac{\sinh kx(\cosh 0)}{\sinh kL} = \frac{\sinh kx}{\sinh kL}$$

$$u(x) = \frac{\sinh kx}{\sinh kL} + p\int_0^{L} G(x,y)\,u(y)\,dy$$

$$u(x) = \frac{\sinh kx}{\sinh kL} + p\int_0^{L} G(x,y)\,\frac{\sinh ky}{\sinh kL}\,dy$$

$$\nabla^2 u + k^2 u = 0$$

$$u|_{r=R} = 0$$

$$u(r,\theta) = 1 - \left(\frac{r}{R}\right)^{\alpha}$$

$$I[u] = \iint (\nabla u)^2 r\,dr\,d\theta$$

$$J[u] = \iint u^2 r\,dr\,d\theta$$

$$I[u] = \int_0^{2\pi}\int_0^{R} \left(\frac{\partial u}{\partial r}\right)^2 r\,dr\,d\theta = 2\pi \int_0^{R} r\left(\frac{\partial u}{\partial r}\right)^2 dr$$

$$J[u] = \int_0^{2\pi}\int_0^{R} u^2 r\,dr\,d\theta = 2\pi \int_0^{R} r u^2\,dr$$

$$\frac{du}{dr} = -\frac{\alpha}{R^{\alpha}} r^{\alpha-1} = -\frac{\alpha}{R}\left(\frac{r}{R}\right)^{\alpha-1}$$

$$I[u] = \int_0^{R} 2\pi r \frac{\alpha^2}{R^2}\left(\frac{r}{R}\right)^{2(\alpha-1)} dr$$

$$= 2\pi \frac{\alpha^2}{R^{2\alpha}} \int_0^{R} r^{2\alpha-1}\,dr = \frac{2\pi\alpha^2}{R^{2\alpha}} \frac{r^{2\alpha}}{2\alpha}\Big|_0^{R} = \frac{2\pi\alpha^2 R^{2\alpha}}{R^{2\alpha}\,2\alpha} = \pi\alpha$$

$$J[u] = 2\pi \int_0^{R} r\left(1 - \left(\frac{r}{R}\right)^{\alpha}\right) dr = 2\pi \int_0^{R} r\left(1 + \left(\frac{r}{R}\right)^{2\alpha} - 2\left(\frac{r}{R}\right)^{\alpha}\right) dr =$$

$$= 2\pi \int_0^{R} r\,dr + 2\pi \int_0^{R} \frac{r^{2\alpha+1}}{R^{2\alpha}} dr - 4\pi \int_0^{R} \frac{r^{\alpha+1}}{R^{\alpha}} dr =$$

$$k[u] = \frac{\frac{1}{\pi}\alpha(\alpha+1)(\alpha+2)}{\frac{1}{\pi}\alpha^2 R^2} = \frac{(\alpha+1)(\alpha+2)}{\alpha R^2}$$

$$\frac{dk}{d\alpha}[u] = \frac{1}{R^2}\frac{[\alpha(\alpha+2)+(\alpha+1)]\alpha - (\alpha+1)(\alpha+2)}{\alpha^2} = \frac{(2\alpha+3)\alpha - \alpha^2 - 3\alpha - 2}{R^2\alpha^2} =$$

$$= \frac{2\alpha^2+3\alpha-\alpha^2-3\alpha-2}{R^2\alpha^2} = \frac{\alpha^2-2}{R^2\alpha^2} = \frac{1}{R^2}\left(1-\frac{2}{\alpha^2}\right) = 0$$

$$1 = \frac{2}{\alpha^2} \iff \alpha^2 = 2$$

~~$0 < \alpha < \sqrt{2}$~~

$\alpha = \sqrt{2}$ pois $\alpha > 0$ logo $\alpha = \pm\sqrt{2}$

$$k(\sqrt{2}) = \frac{\frac{1}{\pi}(\sqrt{2}+1)(\sqrt{2}+2)}{R^2\sqrt{2}}$$

$$k[u(\sqrt{2})] = \frac{2+3\sqrt{2}+2}{R^2\sqrt{2}} = \frac{1}{R^2}\left(\frac{2}{\sqrt{2}}+3+\frac{2}{\sqrt{2}}\right) = \frac{1}{R^2}(2\sqrt{2}+3)$$

$$u(r,\theta) = r(R-r)\cos\theta$$

$$\nabla u = \frac{d}{dr}(rR-r^2)\cos\theta\,\hat{r} - \frac{1}{r}r(R-r)\sin\theta\,\hat{\theta}$$

$$\nabla u = (R-2r)\cos\theta\,\hat{r} - (R-r)\sin\theta\,\hat{\theta}$$

$$(\nabla u)^2 = (R-2r)^2\cos^2\theta + (R-r)^2\sin^2\theta$$

$$I[u] = \int_0^{2\pi}\int_0^R \pi\left[(R-2r)^2\cos^2\theta + (R-r)^2\sin^2\theta\right] dr\,d\theta =$$

$$= \int_0^R \left(r(R-2r)^2\pi + r(R-r)^2\pi\right) dr =$$

$$= \pi\int_0^R \left(r(R^2-4rR+4r^2) + r(R^2+r^2-2Rr)\right) dr =$$

$$= \pi\left[R^2\frac{r^2}{2} - 4R\frac{r^3}{3} + 4\frac{r^4}{4} + R^2\frac{r^2}{2} + \frac{r^4}{4} - 2R\frac{r^3}{3}\right]\Big|_0^R =$$

$$= \pi\left[\frac{R^4}{2} - \frac{4R^4}{3} + R^4 + \frac{R^4}{4} + \frac{R^4}{4} - \frac{2R^4}{3}\right] =$$

$$= \pi R^4\left(\frac{1}{2} - \frac{4}{3} + 1 + \frac{1}{4} + \frac{1}{4} - \frac{2}{3}\right) = \pi R^4\left(2 - \frac{6}{3}\right) = \pi R^4(2-2) = 0$$

$$\lim_{x\to\pm\infty}\psi(x)=e^{ikx}+A_{\pm}e^{\pm ikx}$$

נ.ז.ל.

כעת נחשב את $A_{\pm}$ עבור $\psi_{(x)}$ כללי:

$$\psi_{(x)}=\psi_{\text{אבה}}\pm\int_{-\infty}^{\infty}G(x,y)\tilde{V}(y)\psi(y)dy$$

נשים לב כי e^{ikx} הוא פתרון $\leftarrow$ אבה הומוגנית

$$\lim_{x\to\pm\infty}\psi(x)=e^{ikx}\pm e^{ikx}\int_{-\infty}^{\infty}\frac{1}{2ik}e^{-iky}\tilde{V}_0e^{-y^2}\psi(y)dy$$

$$A_{+}$$

$$\lim_{x\to-\infty}\psi(x)=e^{ikx}\pm e^{-ikx}\int_{-\infty}^{\infty}\frac{1}{2ik}e^{iky}\tilde{V}_0e^{-y^2}\psi(y)dy$$

$$A_{-}$$

$$\psi(x)=e^{ikx}\pm\int_{-\infty}^{\infty}G(x,y)\tilde{V}_0e^{-y^2}e^{iky}dy$$

נחשב את A_{-} עבור הפוטנציאל הנתון:

$$A_{-}^{Born}=\int_{-\infty}^{\infty}\frac{1}{2ik}\tilde{V}_0e^{iky}e^{-y^2}e^{iky}dy=\int_{-\infty}^{\infty}\frac{1}{2ik}\tilde{V}_0e^{-y^2}dy=(*)$$

$$-y^2+2iky-k^2+k^2=-(y-ik)^2-k^2$$

$$(*)=\int_{-\infty}^{\infty}\frac{1}{2ik}\tilde{V}_0e^{-k^2}e^{-(y-ik)^2}dy$$

$$\lambda=y-ik\qquad d\lambda=dy$$

$$A_{-}^{Born}=\frac{\tilde{V}_0e^{-k^2}}{2ik}\int_{-\infty}^{\infty}e^{-\lambda^2}d\lambda=\frac{\tilde{V}_0e^{-k^2}}{2ik}\sqrt{\pi}$$

$$R^{Born} = |A_-^{Born}|^2 = A_- A_-^* = \tilde{V}_0^2 e^{-2k^2} \pi \frac{1}{2ik} \frac{1}{-2ik}$$

$$R^{Born} = \frac{\tilde{V}_0^2 \pi e^{-2k^2}}{4k^2} = \frac{V_0^2 \pi e^{-2k^2}}{4k^2} \frac{4m^2}{\hbar^4}$$